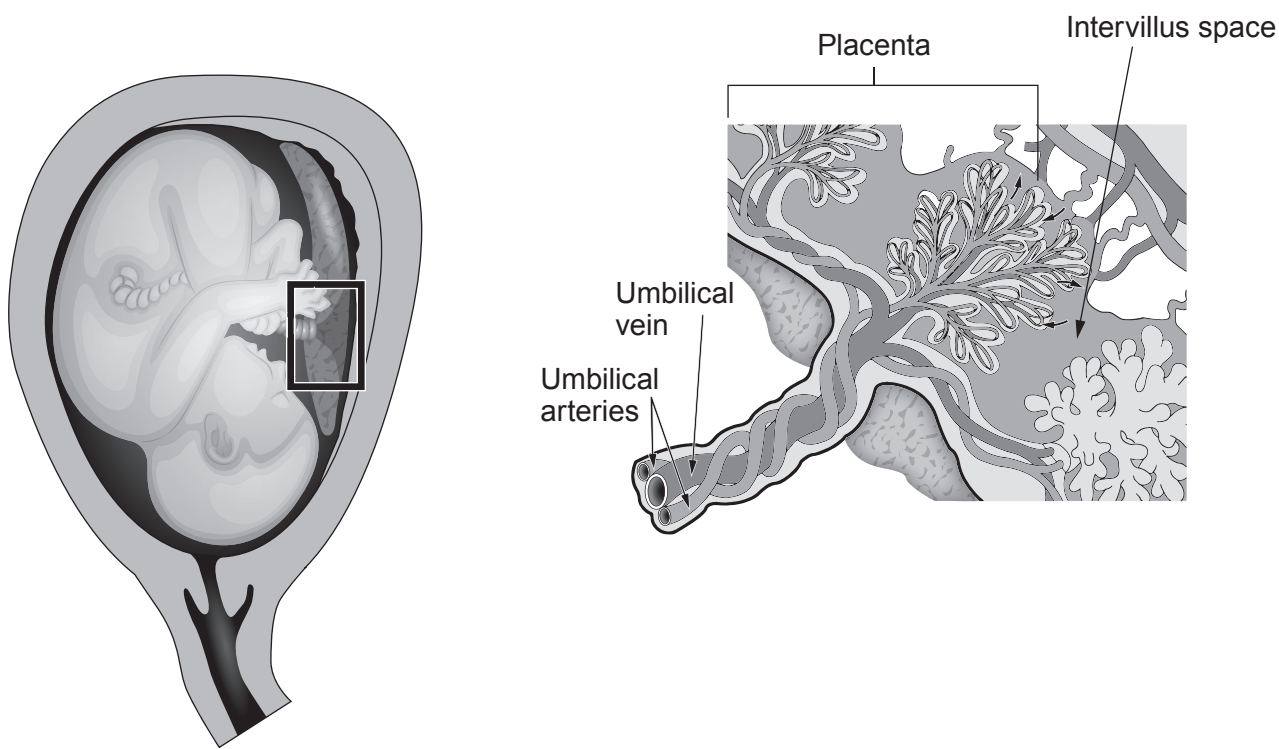


4. The diagram below shows the structure of a human placenta.



(a) State **four** differences between the composition of the blood in the umbilical arteries and the umbilical vein. [2]

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(b) Give **two** reasons why there must be a barrier between foetal and maternal blood systems. [2]

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(c) Suggest how the following features of the placenta enable it to carry out its function.

(i) The arterial blood flow to the placenta is high ($700\text{ cm}^3\text{ min}^{-1}$) and the blood in the intervillous space is exchanged three times per minute. [1]

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(ii) The pressure in the uterine arteries is ten times greater than the pressure in the intervillous space. [1]

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(iii) The length of the capillaries in the placenta is about 320 km. [1]

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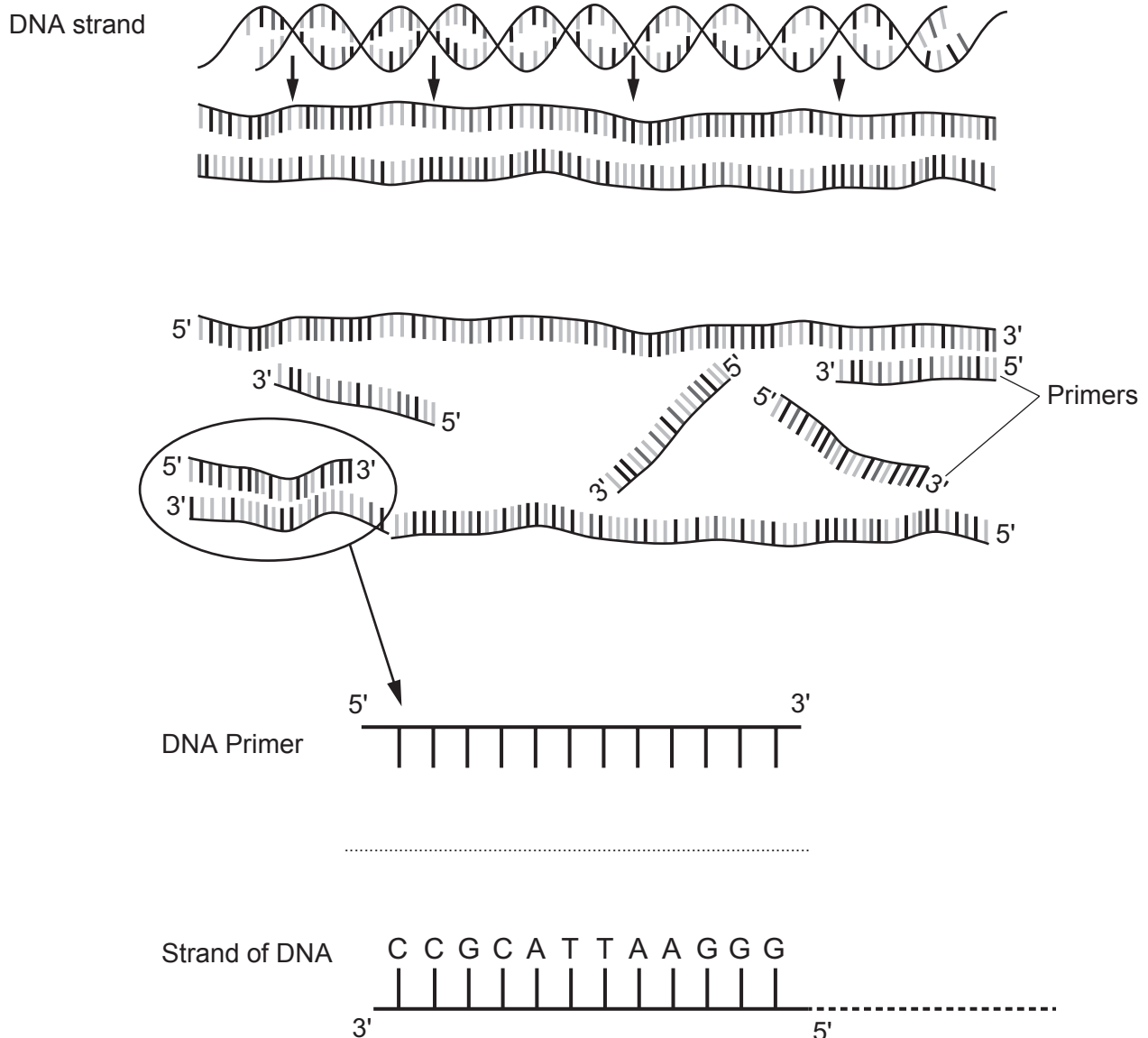
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- (d) During pregnancy, foetal cells die and release many different fragments of DNA into the mother's blood. A blood sample can be taken from the mother and foetal DNA fragments isolated and replicated (amplified) using the polymerase chain reaction (PCR). Many chromosomal abnormalities can be identified using this DNA, by a technique called non-invasive prenatal diagnosis.

Two different primers are needed in the polymerase chain reaction as shown in the diagram below.



- (i) **On the diagram above complete** the nucleotide sequence of **the DNA primer** complementary to the strand of DNA. [1]



(ii) Explain why two different DNA primers are required. [1]

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(iii) A sample of blood from a pregnant woman is taken. Primers specific to a gene on chromosome 21 are used to replicate it. Primers specific to another gene on a different chromosome are used at the same time. These act as a control. Both primers have fluorescent markers attached.

Suggest why it is important to use primers which are specific to a certain gene on each chromosome. [1]

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(iv) After the PCR, the DNA fragments are separated using gel electrophoresis and the level of fluorescence for each gene is measured. The level of fluorescence corresponds to the quantity of that gene.
The quantity of both genes is expressed in the form of a ratio as shown below.

Quantity of gene from chromosome 21 : Quantity of gene from control chromosome.

Suggest why it is necessary to express the quantity of the genes as a ratio. [2]

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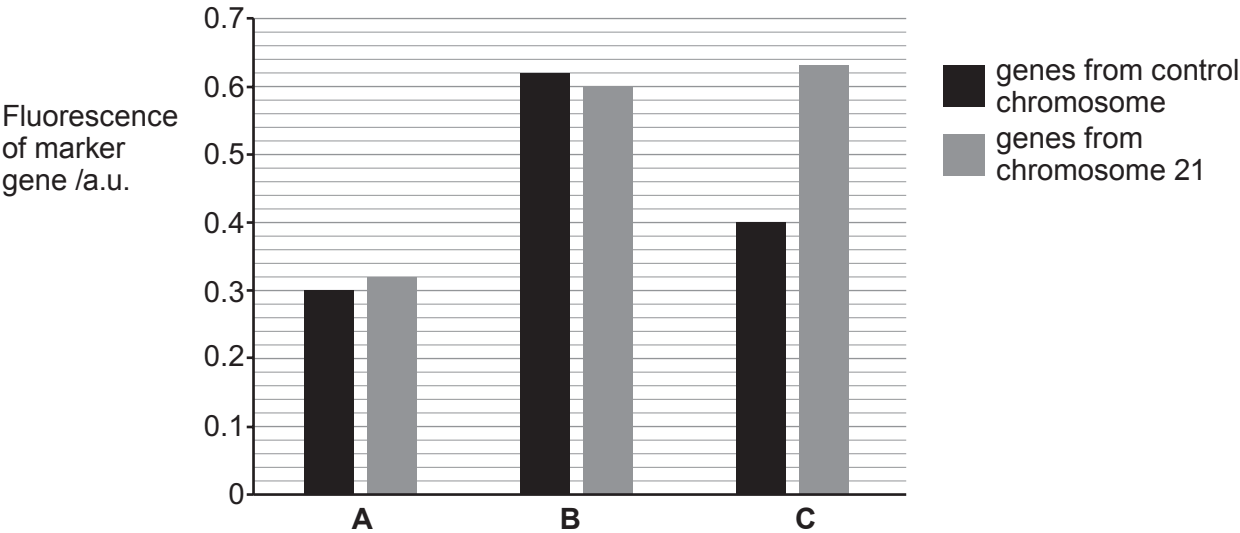
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(v) The diagram below shows the quantity of genes from chromosome 21 and the quantity of the control gene following PCR in three pregnant women **A**, **B** and **C**.



With reference to the ratio of the genes, state and explain the conclusions you can make from the results for **A**, **B** and **C**. [4]

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(e) Suggest **two** concerns that some people may have about the ethics of prenatal diagnosis. [2]

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